

Cost-Volume-Profit (CVP) Analysis

1. Contribution Margin (CM)

CVP analysis is a powerful tool used by managers to understand the relationships between costs, volume (activity), and profit. It helps in making key decisions about pricing, product mix, and cost structure.

The Contribution Margin (CM) is the amount remaining from sales revenue after variable expenses have been deducted. It represents the amount available to cover fixed expenses and then provide profit.

- Contribution Margin (in Total):

$$\text{CM} = \text{Sales Revenue} - \text{Variable Expenses}$$

- Contribution Margin (Per Unit):

$$\text{CM per Unit} = \text{Selling Price per Unit} - \text{Variable Expense per Unit}$$

1. Contribution Margin (CM) Ratio

The CM Ratio (or Contribution Margin Percentage) is the contribution margin expressed as a percentage of total sales revenue. It indicates the percentage of each sales dollar that contributes toward covering fixed expenses and generating profit.

CM Ratio:

CM Ratio=

Total Contribution Margin / Total Sales

OR

CM Ratio=

CM per Unit / Selling Price per Unit

The CM ratio is particularly useful for quickly estimating the effect on profit from a given change in sales dollars. A 25% CM ratio means that for every additional dollar of sales, net operating income will increase by \$0.25 (assuming fixed costs don't change).

2. Break Even Analysis

The Break-Even Point (BEP) is the level of sales at which the company's total revenue equals its total expenses, resulting in zero net operating income.

We can calculate the BEP in units or sales dollars using two common methods:

A. Equation Method (or Formula Method Derivation)

The basic profit equation is:

$$\text{Profit} = (\text{Sales} - \text{Variable Expenses}) - \text{Fixed Expenses}$$

$$\text{Profit} = (\text{Selling Price} \times Q) - (\text{Variable Expense per Unit} \times Q) - \text{Fixed Expenses}$$

Where Q is the quantity of units sold. At the break-even point, Profit=\$0.

$$\$0 = (\text{CM per Unit} \times Q) - \text{Fixed Expenses}$$

2. Break Even Analysis (Cont'd)

B. Formula Method

Break-Even Point in Units: This tells you the number of units to sell to break even.

BEP (Units)=

Total Fixed Expenses/CM per Unit

Break-Even Point in Sales Dollars: This tells you the total sales revenue needed to break even.

BEP (Dollars)=

Total Fixed Expenses/CM Ratio

3. Target Profit Analysis

Target Profit Analysis is an extension of break-even analysis used to determine the sales level (in units or dollars) needed to achieve a specific profit goal (the target profit).

Units to Achieve Target Profit:

$$\text{Units} = \frac{\text{Total Fixed Expenses} + \text{Target Profit}}{\text{CM per Unit}}$$

Sales Dollars to Achieve Target Profit:

$$\text{Sales Dollars} = \frac{\text{Total Fixed Expenses} + \text{Target Profit}}{\text{CM Ratio}}$$

4. Margin of Safety (MOS)

The Margin of Safety (MOS) is the excess of budgeted (or actual) sales over the break-even sales level. It indicates how far sales can drop before the company incurs a loss.

MOS in Dollars:

$$\text{MOS (Dollars)} = \text{Total Budgeted Sales} - \text{Break-Even Sales}$$

MOS Percentage:

$$\text{MOS Percentage} = \text{Margin of Safety in Dollars} / \text{Total Budgeted Sales}$$

5. Degree of Operating Leverage (DOL)

Operating Leverage is a measure of how sensitive net operating income is to percentage changes in sales. Companies with high fixed costs relative to variable costs have high operating leverage.

The Degree of Operating Leverage (DOL) is calculated at a specific level of sales:

$$\text{DOL} = \text{Contribution Margin} / \text{Net Operating Income}$$

The DOL is used to predict the percentage change in net operating income for a given percentage change in sales:

$$\% \Delta \text{ in Net Operating Income} = \text{DOL} \times (\% \Delta \text{ in Sales})$$

Note: DOL is not constant; it is highest near the break-even point and decreases as sales and profits rise.

6. Contribution Format Income Statement

This statement separates costs into variable and fixed categories, which is essential for CVP analysis. It is the format used in this problem set.

Item	Calculation
Sales	Price X Units Sold
- Variable Expenses	Variable Cost X Units Sold
= Contribution Margin	Sales - Variable Expenses
- Fixed Expenses	Total Fixed Costs
= Net Operating Income	Contribution Margin - Fixed Expenses

Problem- CVP Relationships

Voltar Company manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below:

	Total	Per Unit	Percent of Sales
Sales (20,000 units)	\$1,200,000	\$60	100%
Variable expenses	<u>900,000</u>	<u>45</u>	<u>? %</u>
Contribution margin	300,000	<u>\$15</u>	<u>? %</u>
Fixed expenses	<u>240,000</u>		
Net operating income	<u><u>\$ 60,000</u></u>		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.

Requirements

1. Compute the company's CM ratio and variable expense ratio.
2. Compute the company's break-even point in both unit sales and dollar sales.
3. Assume that sales increase by \$400,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase? Use the CM ratio to compute your answer.
4. Refer to the original data. Assume that next year management wants the company to earn a profit of at least \$90,000. How many units will have to be sold to earn this target profit?
5. Refer to the original data. Compute the company's margin of safety in both dollar and percentage form.

Requirements

6. a. Compute the company's degree of operating leverage at the present level of sales.

b. Assume that through a more intense effort by the sales staff, the company's sales increase by 8% next year. By what percentage would you expect net operating income to increase?

Use the degree of operating leverage to obtain your answer.

c. Verify your answer to (b) by preparing a new contribution format income statement showing an 8% increase in sales.

Requirements

7. Refer to the original data. In an effort to increase sales and profits, management is considering the use of a higher-quality speaker. The higher-quality speaker would increase variable costs by \$3 per unit, but management could eliminate one quality inspector who is paid a salary of \$30,000 per year. The sales manager estimates that the higher-quality speaker would increase annual sales by at least 20%.

a. Assuming that changes are made as described above, prepare a projected contribution format income statement for next year. Show data on a total, per unit, and percentage basis.

b. Compute the company's new break-even point in both unit sales and dollar sales. Use the formula method.

c. Would you recommend that the changes be made?